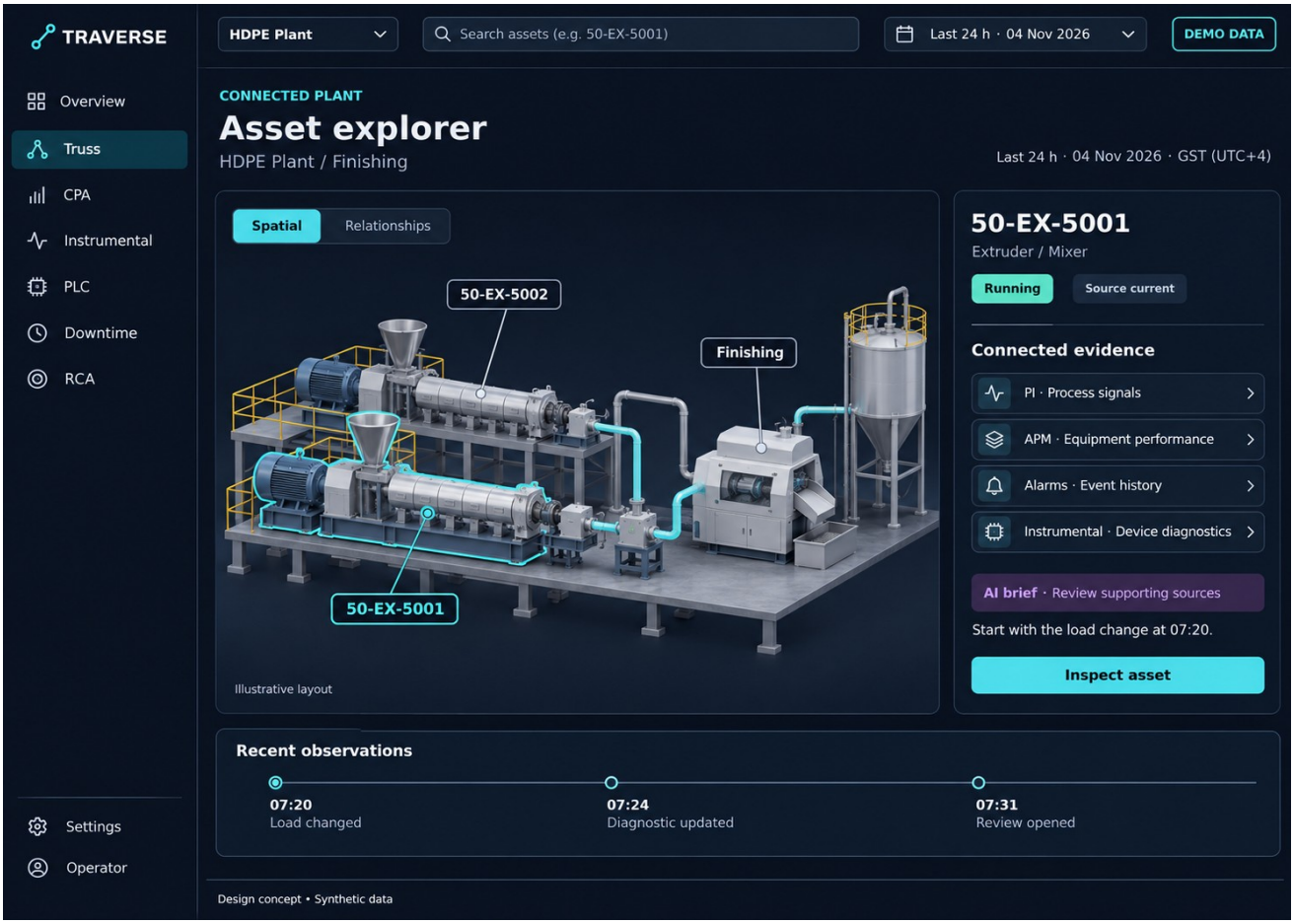
Traverse User Interface Design and Development Guide

Shared design system and module screen specifications

R01 | 04 November 2026 | Prepared for the Traverse development team

Build the Traverse suite on one Graphite and Cyan design system with coordinated Day and Night modes. This guide defines the visual baseline, shared behaviour and twelve page concepts across Truss, CPA, Instrumental, PLC, Downtime and RCA.



Development baseline. Use the exact tokens and written rules in this guide for implementation. The screen concepts establish composition and hierarchy. Page-specific corrections resolve inconsistencies in the generated illustrations.

PLC workflow scope is provisional. All displayed process values and events are demonstration examples. Functional approval and live source validation remain separate from this visual handoff.

Guide map and implementation scope

This is the consolidated R01 handoff for the six modules shown in the concept set. It gives developers one visual reference and a consistent interpretation of each page. Shared rules apply to every module and to future pages added to the suite.

Section	Pages	What to use
Shared foundations	3 to 7	Exact colours, semantics, charts, application shell and data behaviour
Truss	8 to 10	Asset explorer, signal binding and context navigation
CPA	11 to 13	Loop overview, detailed performance and metric handling
Instrumental	14 to 16	Diagnostic fleet, device history and maintenance context
PLC	17 to 19	Provisional production context and record investigation
Downtime	20 to 22	Shift events, event review and duration reconciliation
RCA	23 to 25	Evidence, hypotheses and corrective action verification
Build and acceptance	26 to 29	Shared components, acceptance checks, corrections and decisions
References and files	30	Supporting sources and implementation package contents

Order of precedence

1. Implement the exact semantic colour tokens and shared rules on pages 3 to 7. A screenshot is not the source of an RGB value, spacing measurement or domain rule.
2. Implement each page using its interaction description and module requirements. Apply the corrections on page 28 when the visual concept differs.
3. Use the images to match hierarchy, density and overall appearance. Rebuild text, controls, charts and diagrams as real interface components.

Included and remaining scope

The twelve concepts cover two pages per module. Every page must support both themes, even where only one theme is illustrated. The Overview navigation item is a reserved suite entry; its page is not specified here. Alarm Management, APM, LIMS, energy and other future modules should reuse this foundation when their page scope is defined.

This revision specifies a development baseline. It does not establish a new backend API, approve metric definitions, or make the visual concepts a live operational system. Proposed contracts and open decisions are identified where they affect implementation.

Core colour tokens

Use these opaque sRGB values through semantic tokens. Day is the initial default; persist the user choice across module navigation. A mode change swaps the full token set while preserving labels, selected objects and workflow state.

Token	Day	Night	Use
surface.page	#F3F6FA	#0B1220	Page background
surface.panel	#FFFFFF	#111C2E	Primary content panel
surface.raised	#E8EEF5	#18263B	Secondary surface or field
surface.hover	#E5ECF4	#213149	Hovered neutral row
surface.selected	#DDF3F6	#143540	Selected row or navigation
border.subtle	#D3DCE8	#2A3A52	Decorative divider
border.strong	#75879C	#6F849F	Essential control boundary
text.primary	#152238	#EAF1FA	Headings and primary values
text.secondary	#4B5D73	#A8B7CC	Supporting text
text.muted	#596B81	#8D9FB7	Readable low emphasis text
action.primary	#006B83	#54D6E8	Primary action and selection
action.hover	#00576B	#81E4F0	Primary action hover
action.active	#00485A	#33BACC	Primary action pressed
action.on	#FFFFFF	#0B1220	Text on primary action
focus	#006B83	#7CE8F4	Keyboard focus ring
ai.fg	#6C45C0	#B8A4FF	Labelled AI content
ai.bg	#F0EAFB	#28213F	AI assessment surface

Application rules

Map surface.panel to --traverse-surface-panel and apply the same naming convention to other dotted paths. Set data-theme="day" or data-theme="night" on the themed root. The supplied JSON is a custom schema; it is not a DTCG interchange file.

Do not create per-module brand overrides. Use border.strong when a boundary is needed to identify a control. Use border.subtle for decoration. Preserve the revised Day text.muted value #596B81; the earlier #60738A did not pass every audited neutral pairing. [6]

Status and diagnostic colours

Keep equipment operation, alarm priority, device diagnostics, source quality, workflow state and AI authorship in separate fields. A colour or badge in one field must not silently set another field.

State	Day foreground	Day background	Night foreground	Night background
Danger	#B42332	#FDECEE	#FF7A85	#351E2B
Warning	#8A5700	#FFF3D6	#F5C451	#332A16
Success	#16724A	#E5F5ED	#56D6A1	#163329
Info	#1764C0	#E9F1FC	#7EB1FF	#1B2D46
Unknown	#4B5D73	#E8EEF5	#A8B7CC	#18263B

Use warning for Pending, Needs review and Incomplete; info for In progress and Available; unknown for Not checked, Unassigned, Missing and Unavailable. Use success for a recorded verified or completed outcome. A supported hypothesis remains informational until its causal conclusion is verified.

Instrumental diagnostic categories

Code	Label	Fill	Text	Family
F	Failure	#D92D45	#FFFFFF	Red
C	Function check	#F79B4A	#152238	Orange
S	Out of specification	#F7D75C	#152238	Yellow
M	Maintenance request	#79B6FF	#152238	Blue

NE107 identifies Failure F, Function check C, Out of specification S and Maintenance request M. Display the code and label together. The colour families are supported by published diagnostic guidance; the exact hexadecimal values above are Traverse design choices. Source terminology such as Maintenance required may be retained with a documented label alias. [4, 5]

Meaning stays explicit

- Normal, unknown, unavailable and diagnostics disabled are separate states. A missing diagnostic result must never look healthy.
- Alarm priority colours, symbols, acknowledgement and shelving come from the approved site configuration. R01 supplies no automatic P1 or P2 mapping.
- Violet identifies labelled AI-assisted content. It does not express confidence, correctness or approval. Source evidence and human review state remain visible.
- Chart series colours are scoped to the chart legend. Reusing a hue there does not change its meaning elsewhere in the interface.

Charts and accessibility

Series	Day	Night	Rendering rule
1	#006B83	#54D6E8	Direct label plus line or marker distinction
2	#6C45C0	#B8A4FF	Direct label plus line or marker distinction
3	#245EB8	#80ABFF	Direct label plus line or marker distinction
4	#9B3C82	#DFA6DE	Direct label plus line or marker distinction
5	#53657B	#9EAEC2	Direct label plus line or marker distinction
6	#277575	#86D3D1	Direct label plus line or marker distinction

Use surface.panel behind charts. Measured values use a solid line; reference or setpoint uses a labelled dashed line; predicted or estimated values use a distinct labelled pattern. In mixed AI views, reserve violet for the explicitly labelled AI layer.

Chart behaviour

- Use the actual series data to draw every point. Summary cards and chart endpoints must derive from the same calculation, boundary and period.
- Expose units, axis scale, time zone, source, aggregation method and coverage. Synchronize the time cursor across related plots and show a value table for precise reading.
- Break lines over unavailable intervals. Estimates may appear only as a separate labelled series. Do not turn missing samples into zero or continuous measured data.
- Use common scales for before and after comparisons where the measure and unit match. Label unequal observation durations; normalize count comparisons by exposure when appropriate.

Accessibility baseline

Target at least 4.5:1 for ordinary text and 3:1 for necessary graphical or control cues against their adjacent surfaces. Always provide a visible alternative to colour, such as a label, code, shape or line style. These rules follow the referenced WCAG criteria. [1, 2, 3]

The foundation audit records 94 passing opaque colour pairings. Its lowest text ratio is 4.587:1 and lowest non-text ratio is 3.093:1. The audit covers the listed token pairs; it does not establish complete accessibility of an implemented screen. [6]

Implement a 2 px focus outline with a 3 px gap exposing the neutral surface. Keep it unclipped. Provide keyboard access to rows, graphs and timeline events, meaningful control names, readable zoom/reflow and reduced motion. Verify every implemented page in both themes.

Shared application shell

Use one reusable shell across the suite. Plant scope, search, time context and user controls stay in the same locations. The following measurements are the proposed implementation baseline in CSS pixels; they take precedence over small variations in the images.

Element	Baseline	Behaviour
Desktop reference	1440 by 1024	Validate also at 1366 by 768 and 1920 by 1080
Sidebar	220 px expanded; 64 px rail	Fixed module order with readable labels when expanded
Header	64 px high	Plant selector, search, time range and environment indicator
Main content	24 px outer padding	16 px panel gap; vertical scrolling for long content
Panel	12 px radius; 20 px padding	Quiet border; avoid nested decorative cards
Buttons and inputs	40 px minimum height	6 px radius; clear focus and disabled explanation
Table rows	44 px default	40 px compact mode; do not shrink text to fit
Inspector	320 to 400 px	Persistent on wide layouts; drawer on smaller layouts
Type	14 px body; 16 px key text	24 px page title; 18 px section title; 12 px metadata
Spacing scale	4 8 12 16 20 24 32	Use shared spacing tokens throughout

Canonical navigation and typography

Use the order Overview, Truss, CPA, Instrumental, PLC, Downtime and RCA. Use one approved Traverse mark and one outlined SVG icon set. Keep icons unchanged between themes. Use the search text Search assets, tags and records. Selected navigation uses surface.selected, action.primary text, a persistent start-edge indicator and text emphasis.

Use a self-hosted UI sans family such as Inter with a system fallback. Use tabular numerals for values and a monospace family for technical identifiers. Keep units adjacent to values and put metric definitions in an accessible help or detail view.

Responsive layout

At widths below 1280 px, collapse the navigation to a rail and move the inspector to a drawer when required. Below 1024 px, use a navigation drawer and stack content in task order. Keep tables in their own scroll regions with key identifiers visible. Provide a list alternative for spatial and graph views. Large display wall layouts require a separate viewing-distance design.

Interaction and data behaviour

Preserve investigation context

Every cross-module link carries canonical plant and asset identifiers, the selected time interval, time zone and any available loop, device, signal, event or case identifier. Returning to the previous page restores its filters, selection and scroll position. Route names are an implementation choice; context preservation is mandatory.

Make data status visible

State	Required presentation
Loading	Show a labelled skeleton or progress state; retain the selected context
Empty	Explain that no matching records exist and offer the relevant filter reset
Partial	Identify missing sources or intervals; label coverage and affected metrics
Stale	Keep the last value with its timestamp and a Stale label
Bad quality	Show the source quality state beside the affected value or interval
Unavailable or error	Keep successful panels visible; offer source-specific retry
Permission restricted	Explain that the requested action or record is restricted

Time and identity contract

Use stable IDs for relationships and joins; treat asset tags and serial numbers as aliases where applicable. The device_id is the device identity when supplied by Instrumental. Record effective installation periods so a historical device query uses the device installed at that time.

Store timestamps with an unambiguous offset or UTC representation, then display the selected site time zone explicitly. The mockups use GST UTC+4 as a demonstration setting. Configure the actual plant time zone and reporting-day boundary independently. Historical event pages must show their pinned event period even if the global range differs.

Actions and AI

Selections update the main view and inspector together. Record edits require validation, a visible save result, reviewer identity and an audit entry. Permission and concurrency checks are enforced by the service. A disabled action has an accessible reason; it must not silently do nothing.

AI panels identify authorship and review state, link to the evidence used and expose missing coverage. They must not invent confidence percentages, source links or verified causes. Operational control writes and alarm acknowledgement require separately defined workflows; the actions in this guide are navigation, investigation and review.

Truss Asset explorer

TRS01 | Illustrated in Night mode | Implement in both Day and Night

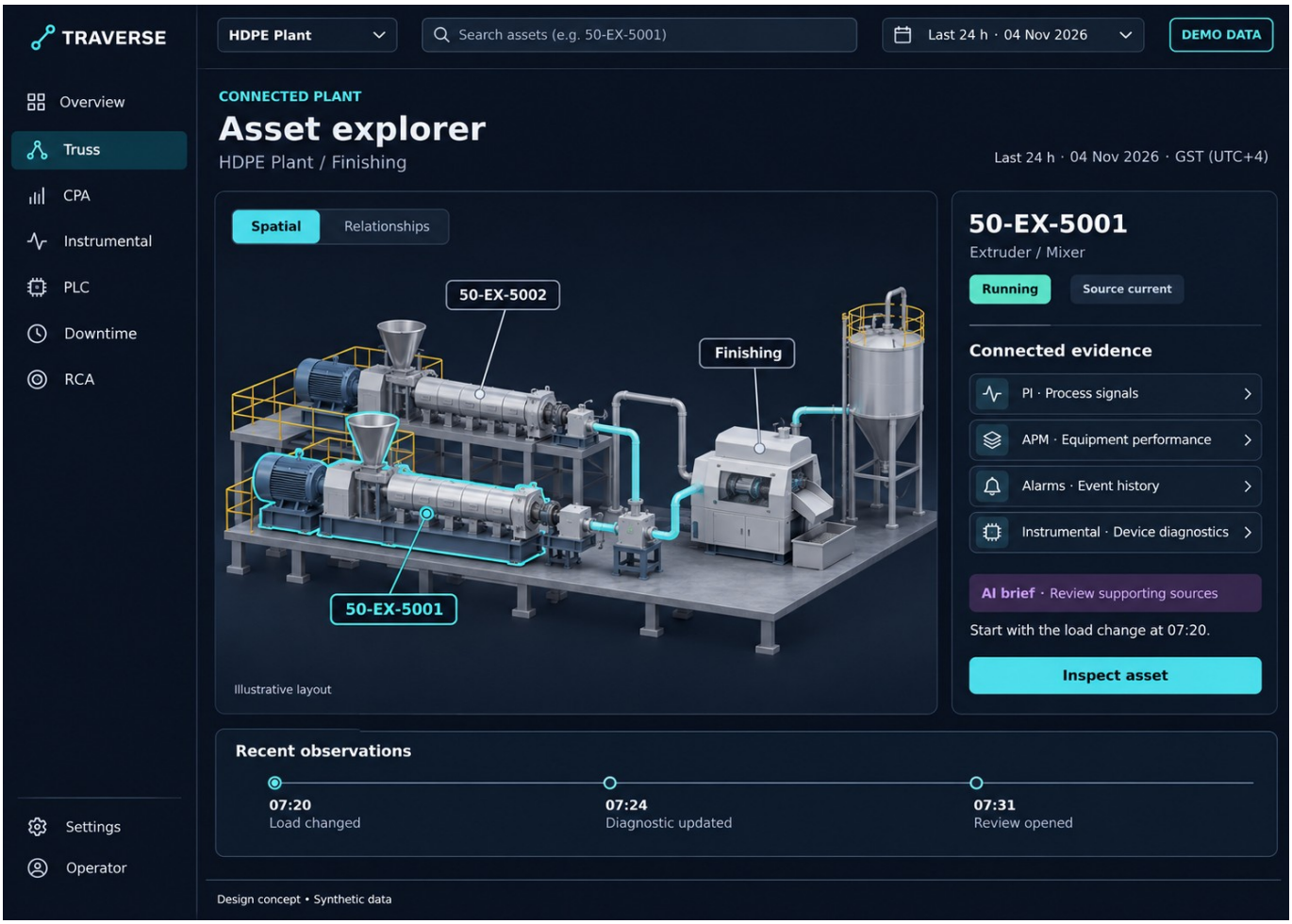


Figure 1. Visual concept. Values, source records and plotted geometry are illustrative.

Purpose. Orient the user within the selected plant and show the asset together with its operating context and available source records.

Interaction. Selecting an asset updates the inspector and observation list. Spatial and relationship views retain the same selected identity. Source links preserve the asset and time range.

Implementation rule. Replace the illustrated 50-EX-5002 object with the validated GearPump_Pelletizer representation. Equipment position, piping and process routes require an authoritative model.

Truss Signal and binding investigation

TRS02 | Illustrated in Day mode | Implement in both Day and Night

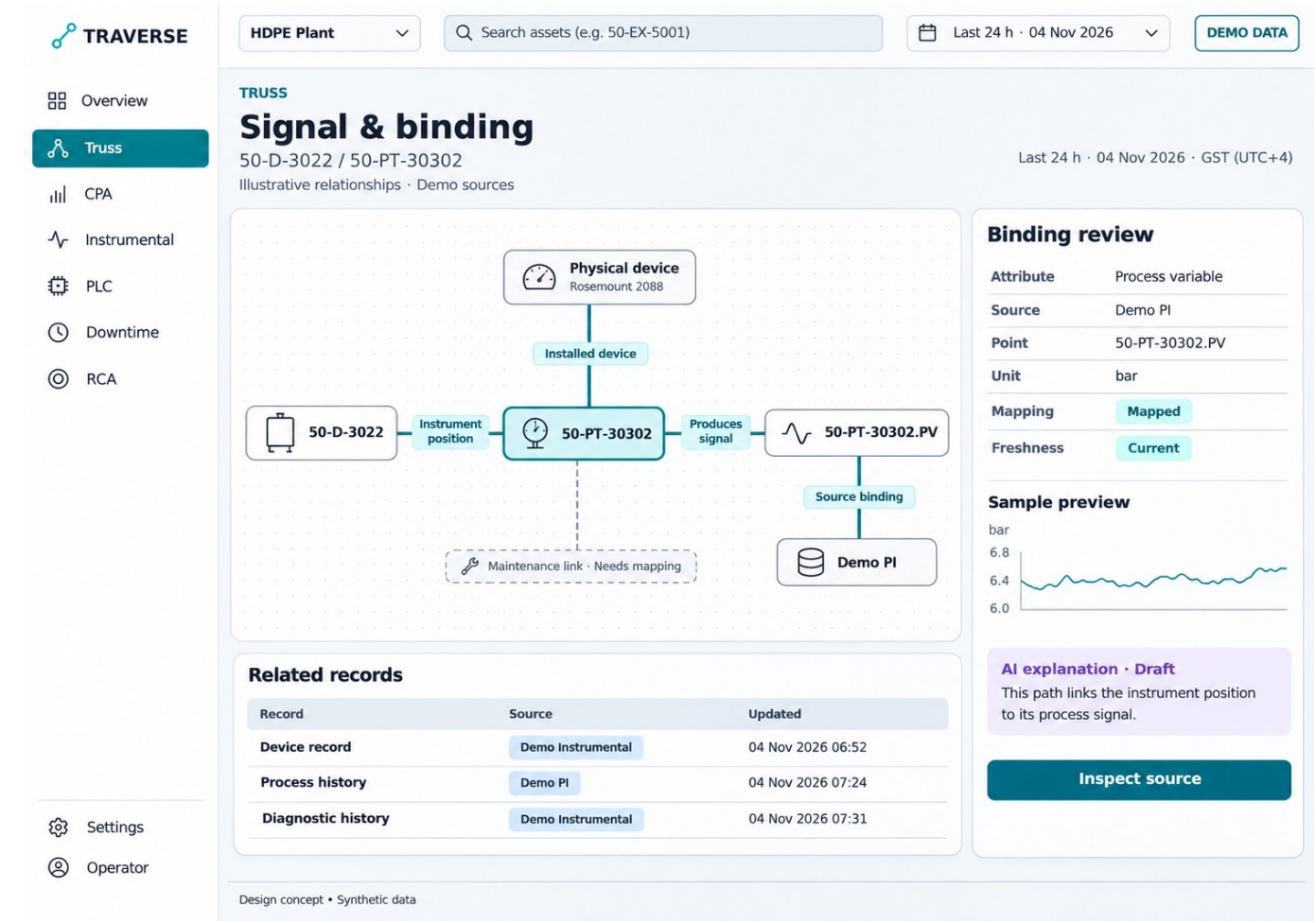


Figure 2. Visual concept. Values, source records and plotted geometry are illustrative.

Purpose. Explain how an equipment position, installed device, signal and source binding relate to the process value being inspected.

Interaction. Selecting a node or edge opens its identity, relationship type and provenance. Selecting the signal exposes its configured source point, unit, last observation and quality.

Implementation rule. Render only verified relationships and bindings as confirmed. An unresolved mapping is visible and cannot open a fictitious source record.

Truss development requirements

Asset explorer

Provide spatial, relationship and list views over the same asset selection. The spatial view is a contextual navigation surface. Show equipment operating state, device condition and telemetry freshness independently. Each observation includes an occurrence time, source and link to supporting evidence.

Keep the selected asset visible when the user changes the time range or moves into CPA, Instrumental, Downtime or RCA. Disable unavailable destinations with a reason such as No linked loop, rather than guessing a target from a similar tag.

Signal and binding investigation

Separate the instrument position from the installed physical device and the measured signal. For the example 50-PT-30302, show the signal 50-PT-30302.PV and its configured historian binding. Display relationship provenance, mapping status and effective dates. Do not use device health as a substitute for process value quality.

Object	Minimum fields for this view
Asset or position	Canonical ID, display tag, name, type, parent and relationship provenance
Physical device	device_id, device tag or serial alias, model, installation period and source
Signal	Signal ID, measured property, engineering unit and producing position
Binding	Source system, point ID, attribute, transformation or scaling, mapping state
Observation	Value, unit, event time, quality, freshness and evidence reference

Required interaction sequence

1. Select 50-EX-5001 or another known asset and open its related signals.
2. Choose a signal, inspect the actual binding and open the associated historical trend.
3. Open a linked diagnostic, control loop or case while retaining the original context.

Asset identity correction

Use 50-EX-5001 as Extruder/Mixer and 50-EX-5002 as GearPump_Pelletizer. The first screen contains illustrative geometry that does not correctly represent this distinction. Relationship data and the maintained asset model govern the final visualization.

CPA Control performance overview

CPA01 | Illustrated in Day mode | Implement in both Day and Night

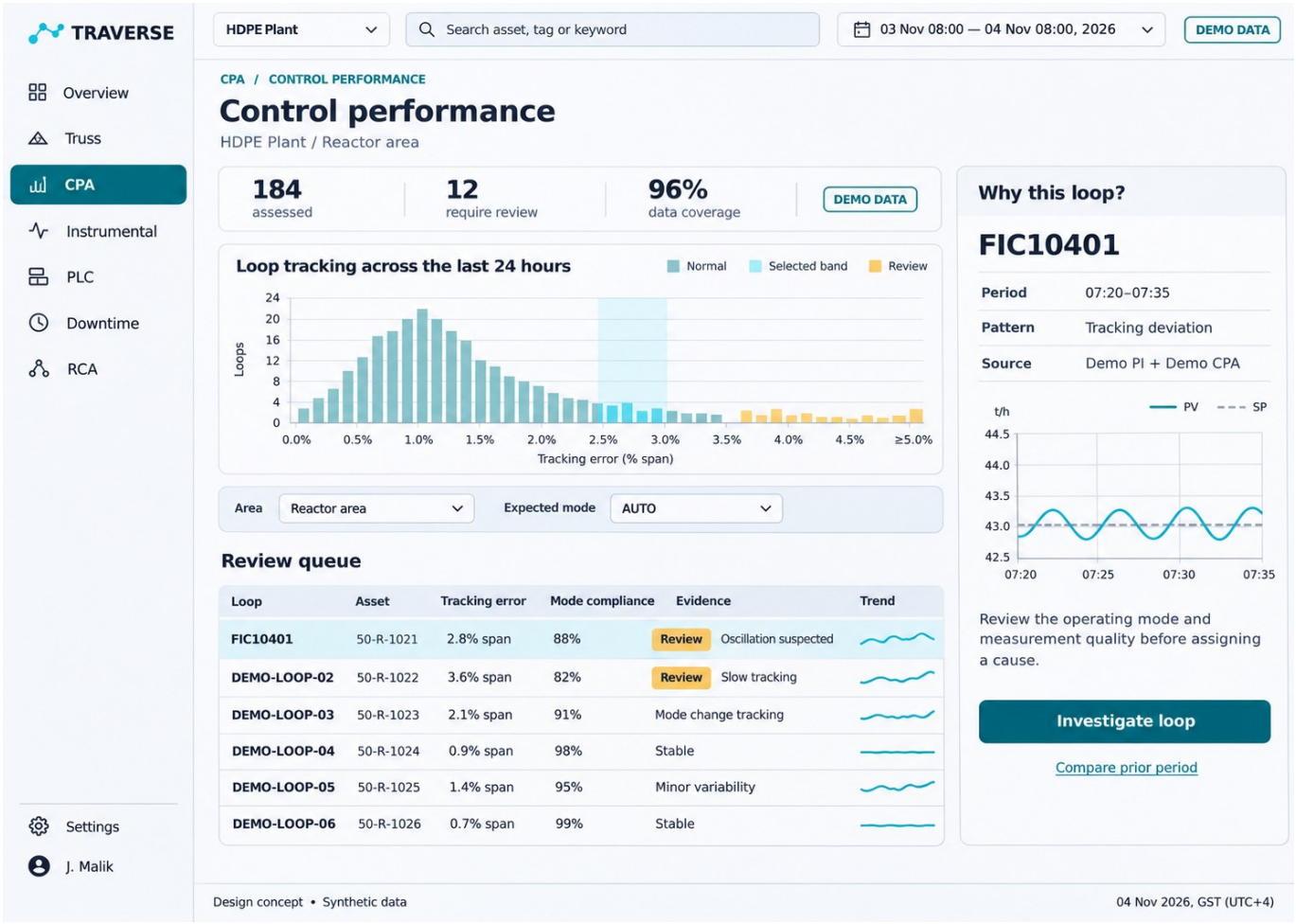


Figure 3. Visual concept. Values, source records and plotted geometry are illustrative.

Purpose. Show the assessed control-loop population and give control engineers a clear queue of loops that need investigation.

Interaction. Filter by plant, area, time window and review condition. Selecting a loop updates its explanation and opens CPA02 with the same interval.

Implementation rule. Show population size, assessment coverage and metric definitions. Keep rows with unavailable metrics visible with an explicit reason; do not rank missing data as good performance.

CPA Loop performance

CPA02 | Illustrated in Night mode | Implement in both Day and Night

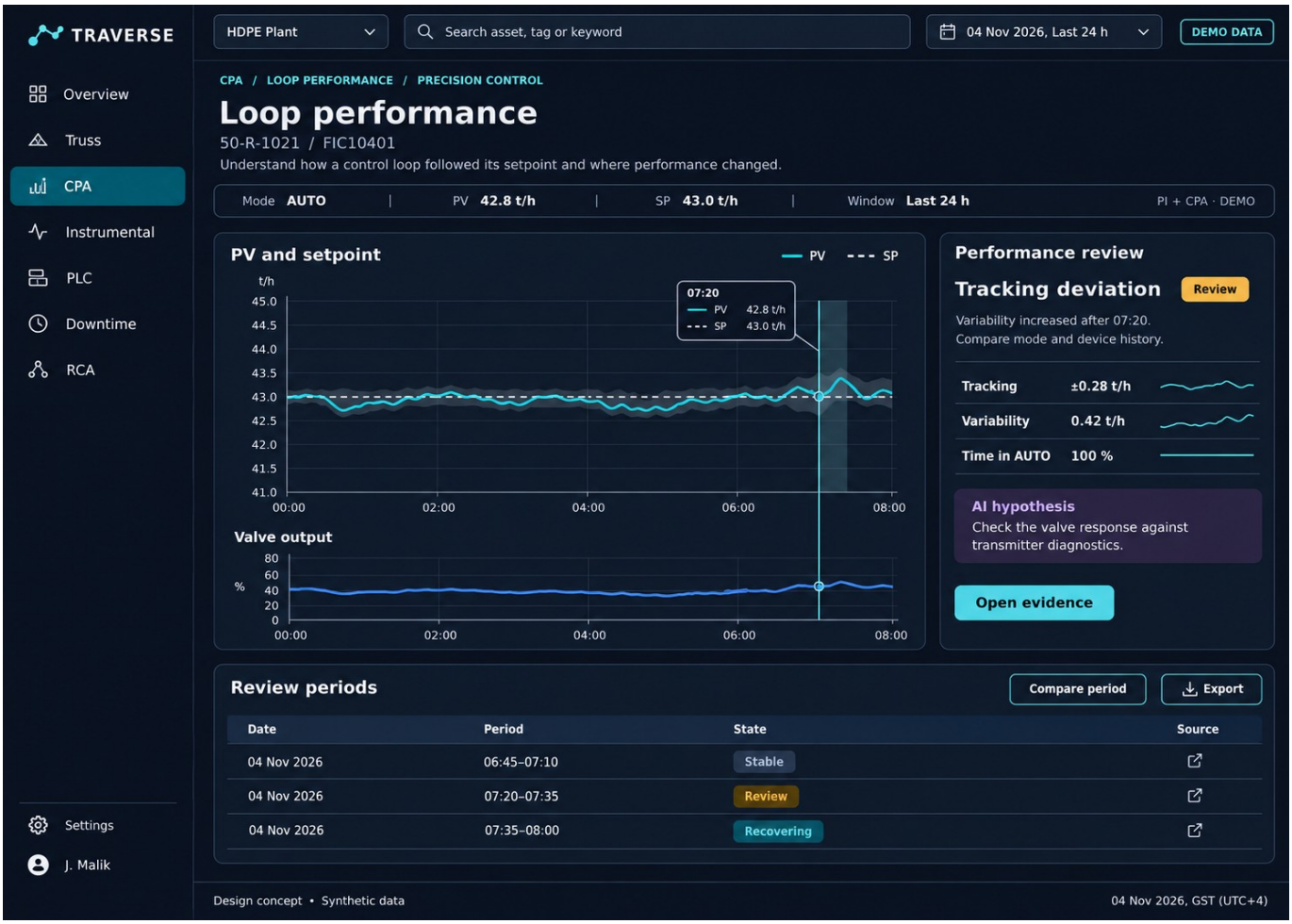


Figure 4. Visual concept. Values, source records and plotted geometry are illustrative.

Purpose. Examine a selected loop using aligned process variable, setpoint, controller output and operating-mode context.

Interaction. Synchronize the cursor and selected interval across charts. Selecting a review period updates the detail panel and evidence links. Related instrument and alarm links retain the loop and period.

Implementation rule. Use the CPA service definition for every metric and threshold. Label output as Controller output unless measured valve position is actually available.

CPA development requirements

Overview and assessment coverage

The overview presents assessed loops, loops needing review and data coverage for the selected population and period. Use a ranked table with loop tag, parent asset, metric value and unit, assessment state, coverage and last evaluation time. Clicking a column heading sorts the eligible rows using a documented rule.

The review explanation states which metric or rule selected the loop. Use clear reasons such as High tracking error or Inadequate data coverage. Keep Unassessed distinct from Meets criterion. Do not combine different engineering units into one unexplained ranking.

Detailed performance

Show PV, SP, controller output and actual mode history on aligned time axes. A valve-position trace is a separate measurement when available. Preserve controller mode changes, manual periods, saturation and bad-quality intervals as context rather than silently excluding them.

Metric calculations come from a versioned service or approved source. Overview and detail use the same definition, evaluation window and denominator; label plot zoom separately. Show unit, eligibility, coverage and baseline. Oscillation, stiction or poor-tuning claims must identify their diagnostic method and evidence; the trend alone is insufficient to certify a cause.

Data group	Display contract
Loop identity	Canonical loop ID, tag, controlled property and related asset
Time series	Timestamped PV, SP, output, mode and quality for the same interval
Metric result	Metric name, value, unit, definition version, coverage and assessment time
Review rule	Threshold or baseline, eligible operating mode and evaluation window
Evidence link	Source record, instrument relation, alarm context or associated case

Review and escalation

Record review captures the selected period, metric snapshot, reviewer and note. Opening an RCA links the existing evidence and creates a traceable relationship; it does not promote a suspected control issue to a verified cause. Source refreshes must not overwrite saved review evidence without versioning.

For FIC10401 on 50-R-1021, demonstrate the path from overview to PV versus SP, mode history, diagnostics and alarms. Define the denominators behind assessed loops, coverage and time in AUTO. A loop labelled Stable needs an explicit review reason. Use a named statistic in place of the ambiguous Tracking plus or minus label; percentage of span requires a valid configured span.

Instrumental Diagnostic fleet

INS01 | Illustrated in Night mode | Implement in both Day and Night

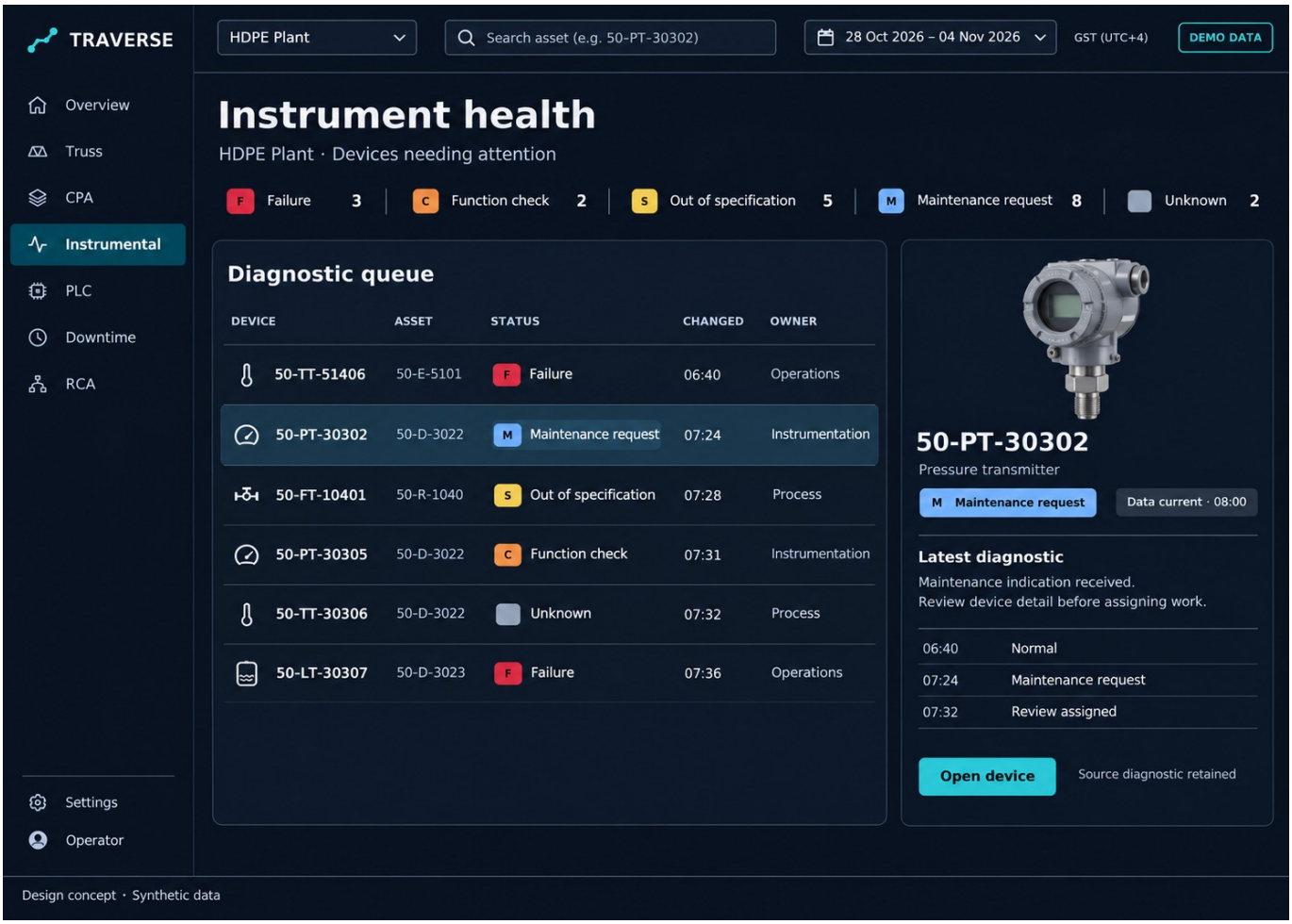


Figure 5. Visual concept. Values, source records and plotted geometry are illustrative.

Purpose. Give the instrumentation team a fleet view of device diagnostic categories and a prioritized review queue.

Interaction. Selecting an F, C, S or M category filters the queue. Selecting a device updates the inspector and opens its history with the current plant and time selection.

Implementation rule. Show diagnostic category, source freshness and data quality independently. Define whether category counters count devices or active diagnostic records, and expose that definition.

Instrumental Device diagnostic history

INS02 | Illustrated in Day mode | Implement in both Day and Night

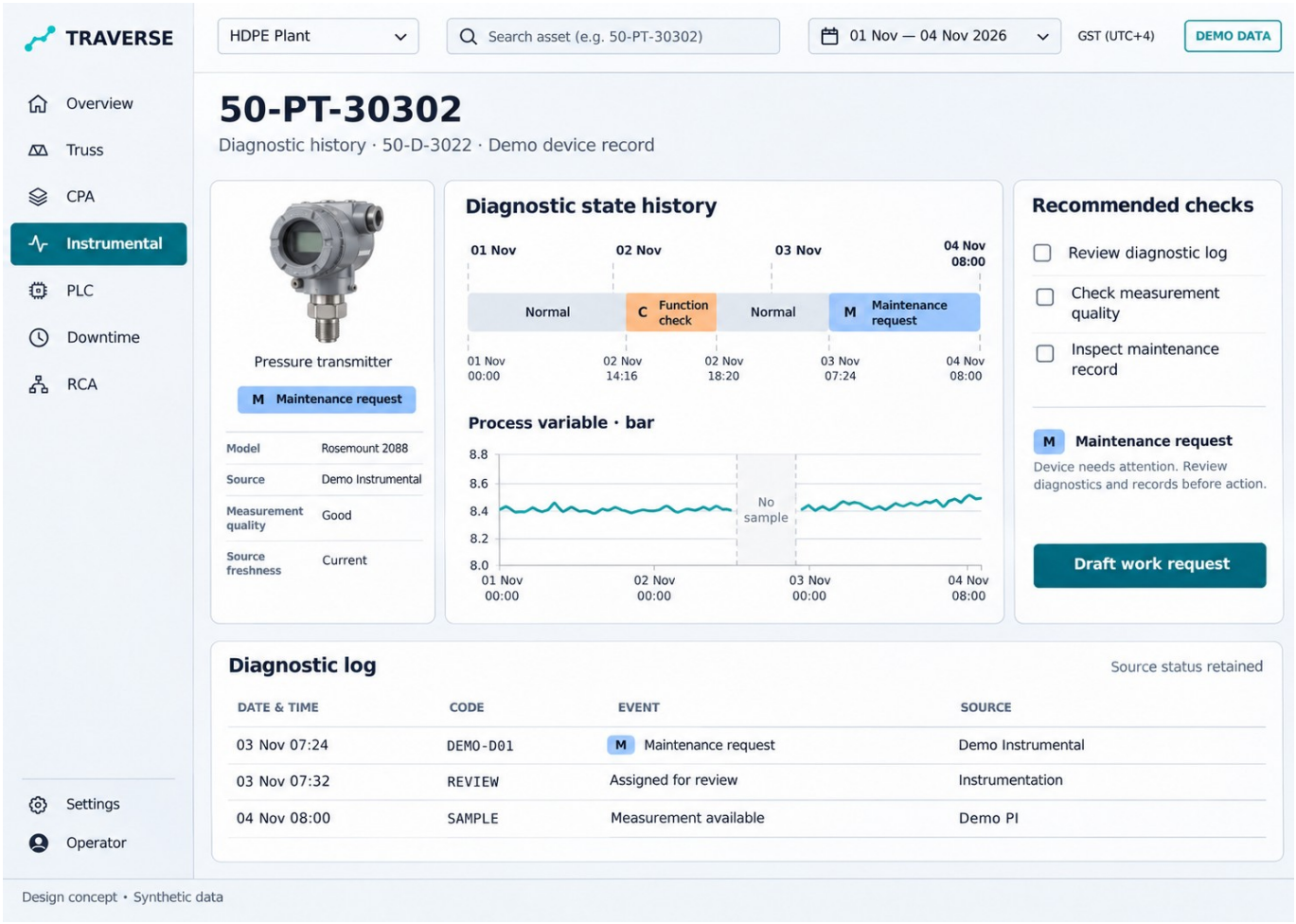


Figure 6. Visual concept. Values, source records and plotted geometry are illustrative.

Purpose. Explain a device diagnostic over time with the related process measurement, diagnostic log and maintenance context.

Interaction. A selected timeline interval filters diagnostic records and aligns the trend. A maintenance action opens a prefilled review or work-request flow only where a configured integration exists.

Implementation rule. Use the installed device identity for the selected historical period. Gaps in diagnostics or PV data remain explicit; a quiet diagnostic history does not prove good device health.

Instrumental development requirements

Fleet queue

Separate the device diagnostic category from the review workflow. A maintenance request M is a diagnostic state; Reviewed is a user workflow state. Review does not clear the diagnostic at its source. A device may also have an unavailable source or stale observation that requires its own indicator.

Define a fleet counter basis before implementation. If counters count active diagnostic records, a device may contribute to more than one category. F, C, S and M are diagnostic categories, not an intrinsic severity scale. If the source reports one summary category, expose its aggregation or precedence rule.

Device history

Use device_id and installation history to identify the physical device. Show diagnostic code, category, text, source time, first seen, last seen and current or cleared state. Include original source messages where available, with the normalized display category alongside them.

Display process PV, device diagnostics and maintenance records on a common time basis. Keep PV quality distinct from diagnostic communication freshness. A device replacement changes the physical-device history without discarding the instrument-position history.

Data group	Required fields
Device identity	device_id, position ID, tag aliases, model and installation period
Diagnostic record	Source code, normalized category, message, active interval and source
Data state	Last received time, source health, freshness threshold and quality state
Review record	Reviewer, time, note, selected evidence and review outcome
Maintenance link	External system, work request or order ID and actual synchronization state

Maintenance workflow

When no CMMS integration is configured, offer a local review note or a clearly labelled request draft. Do not display a successful work order submission until the external system returns an identifier. Retries must preserve a single intended request and expose failures.

Use the F, C, S and M mapping on page 4, including white text on the red F fill. Place Unknown under status availability rather than as a fifth NAMUR category. Render history intervals proportionally to time. If diagnostics, workflow actions and sample-quality events share a table, call it Event log and include an event-type column.

PLC Production context

PLC01 | Illustrated in Night mode | Implement in both Day and Night

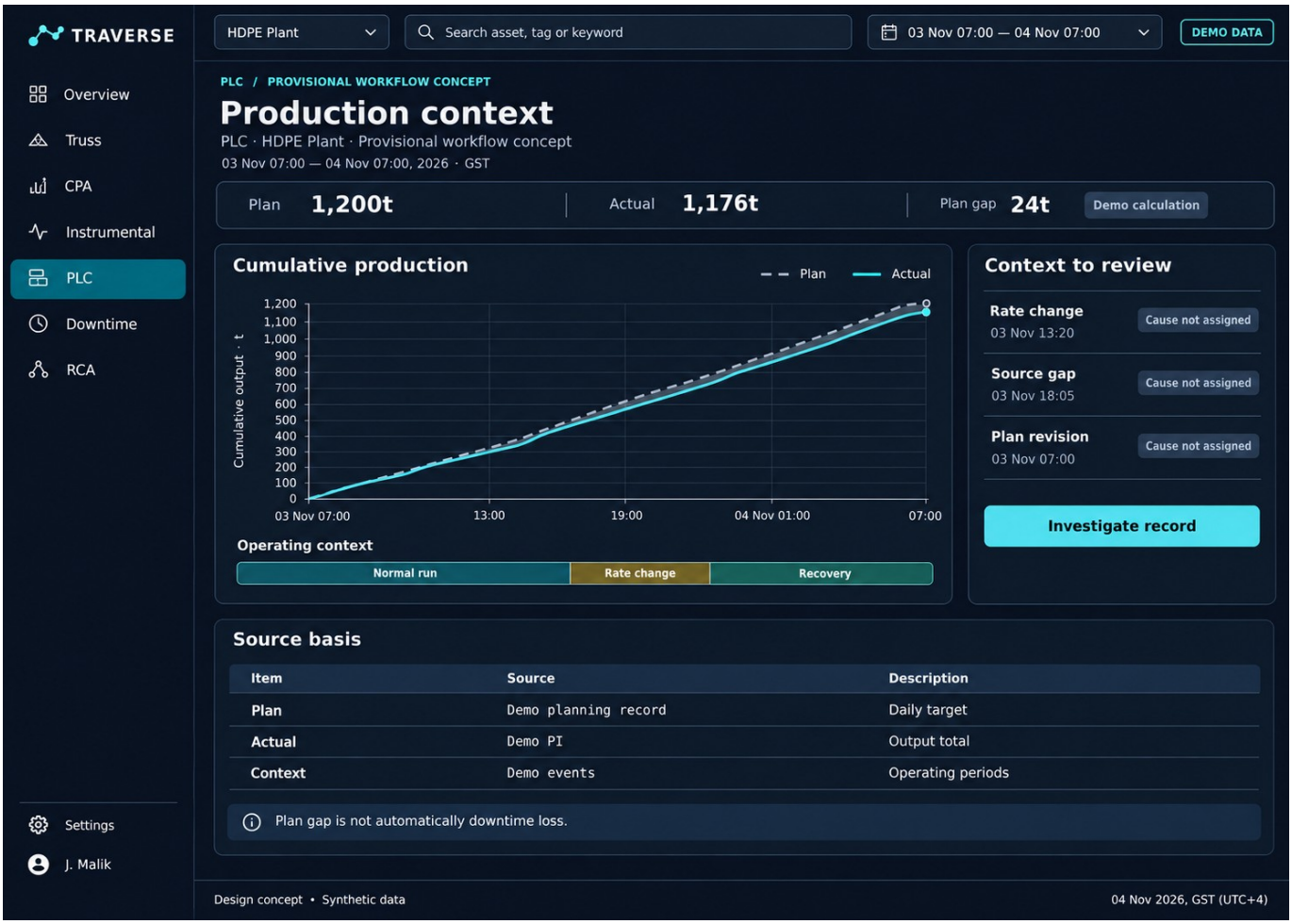


Figure 7. Visual concept. Values, source records and plotted geometry are illustrative.

Purpose. Explore a provisional production-context overview that compares plan and actual and lists records requiring investigation.

Interaction. Changing the reporting period recalculates the displayed plan, actual and gap for the same production boundary. Selecting a context record opens PLC02.

Implementation rule. PLC scope is unconfirmed. The shown 1200 t plan and 1176 t actual yield a 24 t gap; the production chart must be redrawn from matching data. This gap is not automatically accounted production loss.

PLC Record investigation

PLC02 | Illustrated in Day mode | Implement in both Day and Night

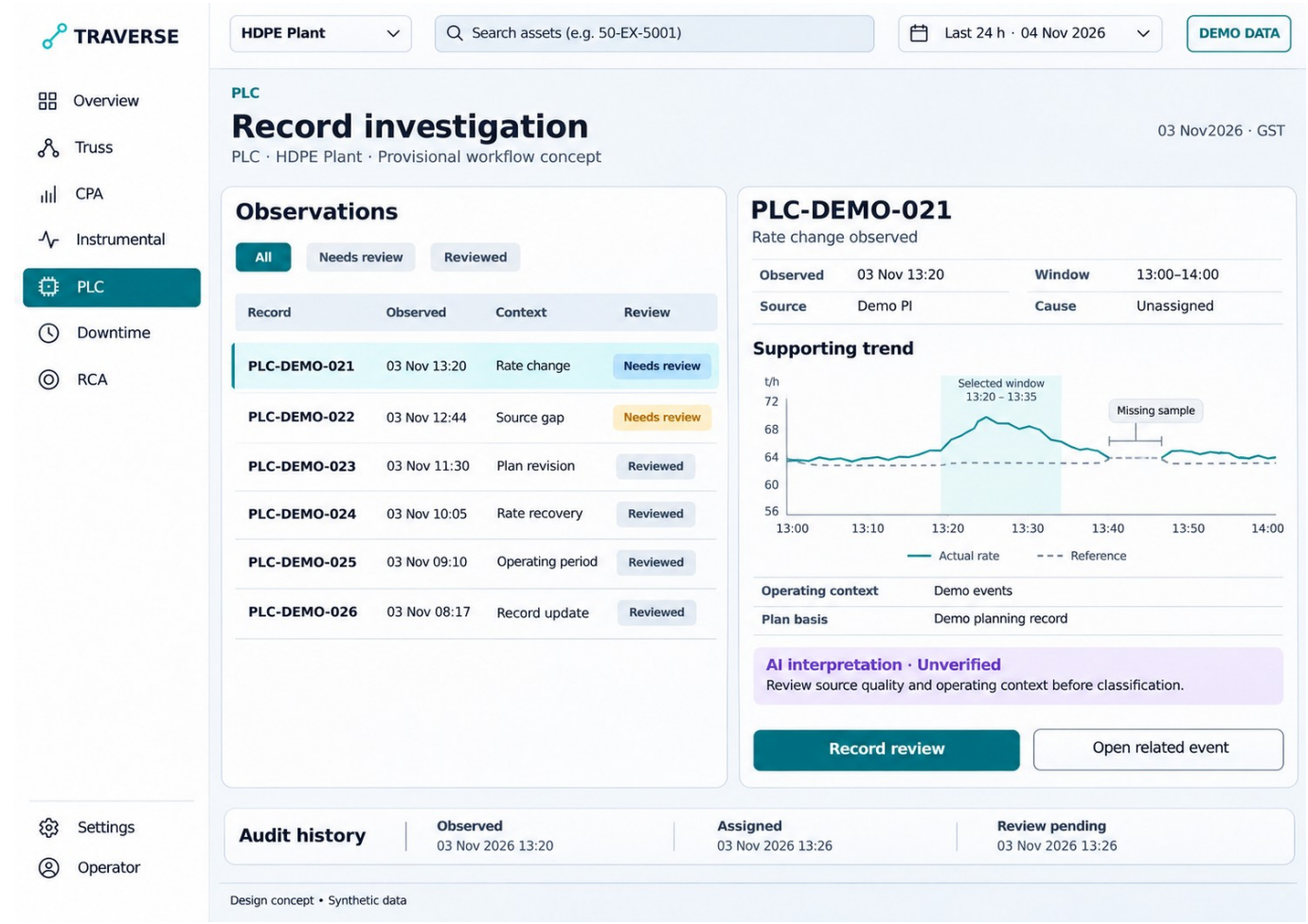


Figure 8. Visual concept. Values, source records and plotted geometry are illustrative.

Purpose. Explore a provisional record review workflow with a ledger, supporting trend and an audit history.

Interaction. Selecting a row updates the record, source references and trend together. Record review captures an outcome and note. Open related event requires an actual linked event.

Implementation rule. Use the same record IDs and timestamps as the overview. Leave missing samples visible. Reviewing a record does not confirm its cause or resolve a source-data gap.

PLC provisional development requirements

Functional scope decision

PLC is retained as the supplied module name. Its expansion, purpose, primary users and owned records require confirmation. These two screens establish a visual pattern only. Do not interpret the name as a programmable logic controller engineering or control interface, or commit the provisional production workflow as approved scope.

Shared shell, theme, table, chart, evidence and review components can be implemented immediately. Keep the provisional page routes and domain data behind a scope decision so the final module can reuse those components without carrying an incorrect business model.

Production context proposal

For the shown concept, compare one plan version with actual production for the same plant or line, product, period and unit. The daily example uses a 07:00 to 07:00 reporting window. Show how actual production was aggregated, what source coverage is available and whether any values are estimated.

The plan gap is plan minus actual for the declared scope. Accounted production loss requires its own event and reconciliation method. A positive gap does not by itself identify downtime, an equipment cause or a recoverable opportunity.

Proposed object	Minimum information
Plan version	Version ID, effective period, product, production boundary, unit and quantity
Actual result	Source, aggregation method, period, quantity, quality and coverage
Context record	Record ID, observed time, type, related asset and evidence links
Review	Reviewer, reviewed time, outcome, note and revision

Record investigation proposal

Allow the user to inspect a rate change, source gap, plan revision or operating period without assigning a cause automatically. Every related event action requires a durable link. Preserve the original observation and append review information as a versioned record.

The overview and record concept currently use different source-gap and plan-revision times. Development must use one coherent fixture or label the records as different observations. A declared source gap must appear in the actual series or in its explicit estimation metadata.

Downtime Shift event timeline

DWT01 | Illustrated in Night mode | Implement in both Day and Night

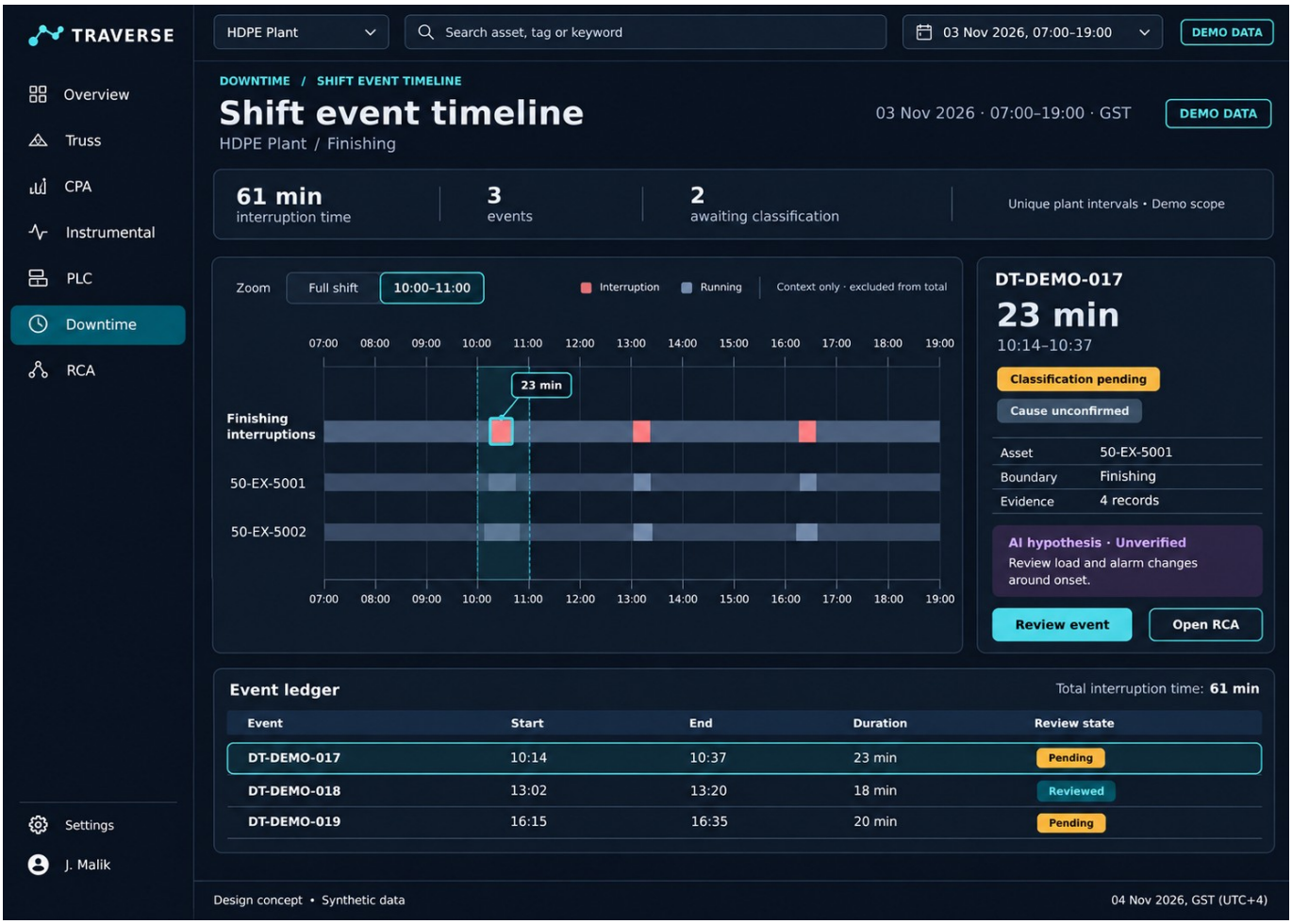


Figure 9. Visual concept. Values, source records and plotted geometry are illustrative.

Purpose. Show interruptions within a shift, their equipment context and the events awaiting review.

Interaction. Timeline, ledger and inspector share one event selection. Review event opens DWT02. Selecting a time window highlights it; Zoom changes the visible time axis.

Implementation rule. Count the union of included plant-level intervals. The example 23, 18 and 20 minute intervals total 61 minutes without overlap. Context lanes must not add the same time again.

Downtime Event review and reconciliation

DWT02 | Illustrated in Day mode | Implement in both Day and Night

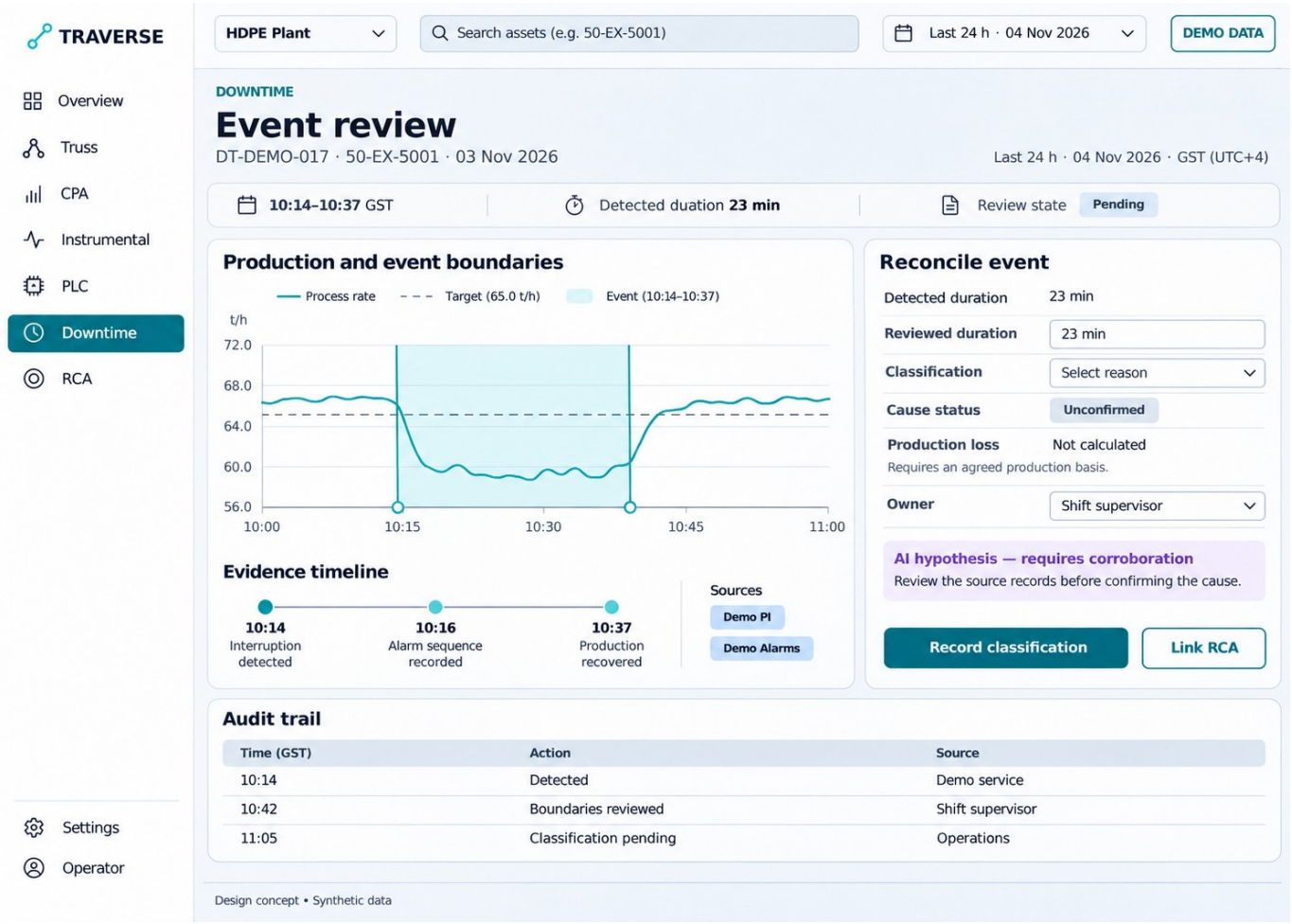


Figure 10. Visual concept. Values, source records and plotted geometry are illustrative.

Purpose. Review an event boundary and classification using production behaviour and timestamped supporting evidence.

Interaction. Authorized reviewers can adjust reviewed boundaries with a reason while preserving detected values. Saving records a version and audit entry. An RCA link remains a separate relationship.

Implementation rule. Define interruption, slowdown and recovery criteria before production use. The 10:37 recovery label in the concept must be reconciled with its plotted trend. Leave loss uncalculated until its basis is available.

Downtime development requirements

Event detection and review

Keep the original detected start and end, the reviewed start and end, detection rule version and review reason. Compute the displayed duration from the selected boundary basis. Record the plant, production boundary, related assets, event type, source and evidence references.

An interruption category may represent a stop or reduced-rate period only if the site definition says so. The illustrated trend stays above zero; it does not demonstrate a full shutdown. Label recovery started, interruption ended or recovered to criterion according to the actual rule used.

Duration and loss accounting

Clip intervals to the selected reporting period, merge overlaps within the declared scope and calculate unique duration. Equipment context lanes explain contributing or coincident events; they do not increase the plant total automatically. Show excluded time and its reason when exclusions apply.

Production loss is a separate result with a defined baseline, product and boundary, eligible interval, actual production calculation and treatment of missing data. Retain measured, estimated and reconciled states. A duration value or plan gap alone must not be converted into tonnes of loss.

Record group	Required fields
Event identity	Event ID, scope, type, related assets and detection-rule version
Boundaries	Detected and reviewed start/end, time zone, duration and revision reason
Classification	Category, review state, cause status, owner and supporting evidence
Loss result	Value, unit, method, baseline, coverage and reconciliation state
Audit	Version, actor, action, timestamp, before/after values and linked case

Review outcome

Record classification saves the reviewed event. It does not certify a root cause. Link RCA opens an existing case or explicitly creates a new one. Preserve historical evidence references so later source updates do not silently change the basis of an approved review.

Use one fixture for the three example events: 10:14 to 10:37, 13:02 to 13:20 and 16:15 to 16:35 on 03 November 2026. Label the event date explicitly when navigating from a global last-24-hour view.

RCA Investigation workspace

RCA01 | Illustrated in Day mode | Implement in both Day and Night

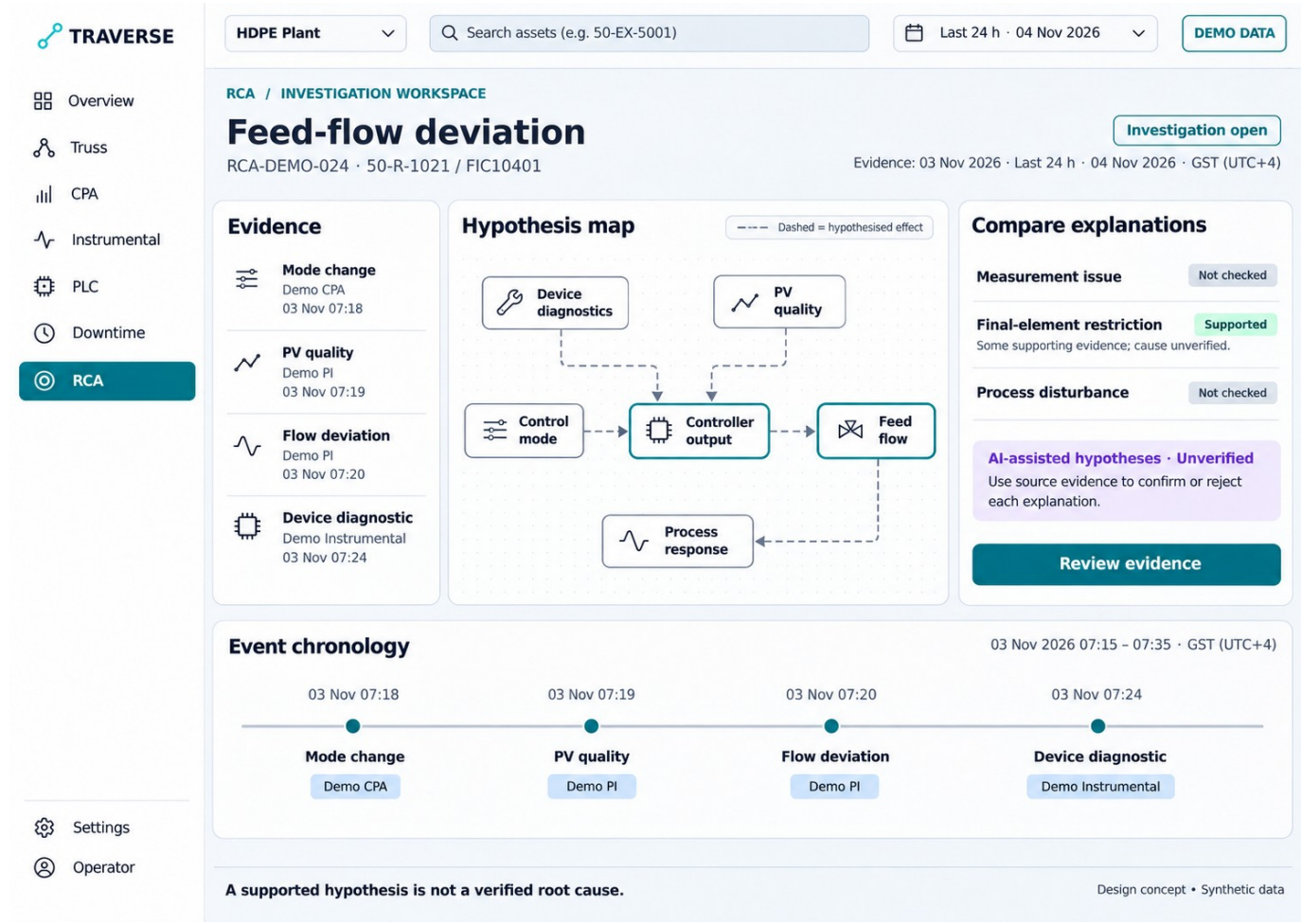


Figure 11. Visual concept. Values, source records and plotted geometry are illustrative.

Purpose. Bring evidence, candidate explanations and chronology into one investigation workspace.

Interaction. Selecting evidence opens its source record and time interval. Selecting a hypothesis shows supporting and contradicting evidence, assumptions and checks still required.

Implementation rule. Dashed relationships in this concept are hypotheses. Supported remains distinct from Verified cause. Use an informational state for partial support and retain the explicit unverified label.

RCA Corrective action verification

RCA02 | Illustrated in Night mode | Implement in both Day and Night

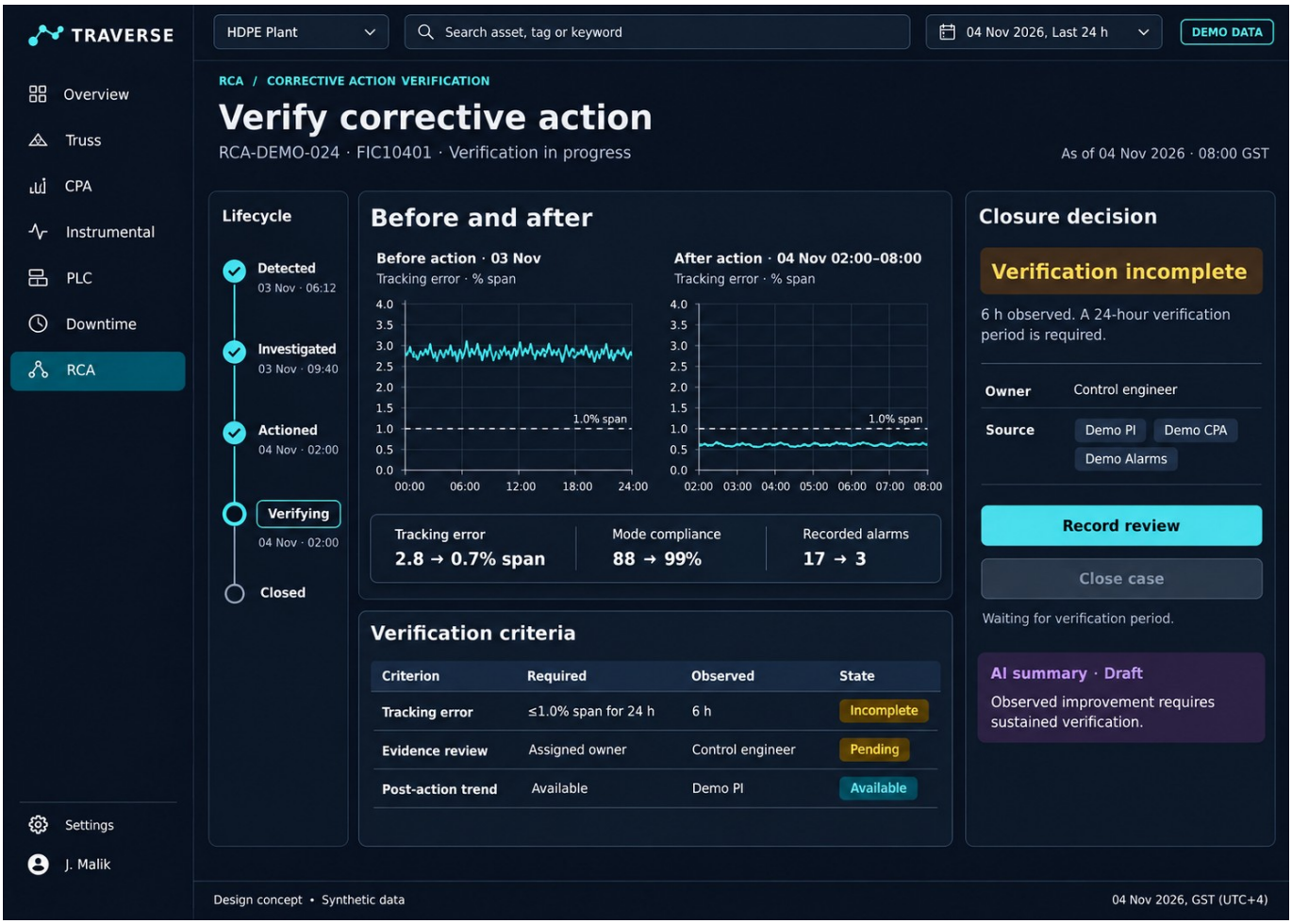


Figure 12. Visual concept. Values, source records and plotted geometry are illustrative.

Purpose. Check whether a corrective action has met its agreed verification criteria before allowing case closure.

Interaction. Show action time, observation period, before/after measures and criterion states. Record review appends evidence. Close case becomes available only after the configured closure rules are satisfied.

Implementation rule. The example has six hours observed against a 24-hour criterion, so verification remains incomplete. Compare alarm rates over comparable exposure; raw counts from unequal durations cannot establish the same improvement.

RCA development requirements

Evidence and hypotheses

Each evidence item identifies its source record, asset or signal, observation period, quality and retrieval time. A hypothesis records its description, author, assumptions, supporting evidence, contradicting evidence and review state. The interface must expose why a relationship is hypothesized, supported, rejected or verified.

Keep observed chronology separate from a causal graph. An event occurring first does not establish it as the cause. Model-generated explanations stay labelled AI-assisted and Unverified until a reviewer applies the agreed investigation method.

Case lifecycle

Use the illustrated lifecycle Detected, Investigated, Actioned, Verifying and Closed as the proposed display sequence. The service owns allowed transitions, roles and prerequisites. Provide a controlled reopen action with a reason when recurrence or new evidence changes the conclusion.

Every corrective action includes an owner, action time, intended effect and verification criteria. Verification specifies the metric, definition version, eligible operating conditions, required duration or sample coverage and acceptance threshold. Store the evidence supporting each criterion result.

Record	Minimum fields
Case	Case ID, title, scope, owner, state, version and related events
Hypothesis	Claim, status, assumptions, author and evidence references
Action	Description, owner, completion time and expected result
Verification	Metric, threshold, eligibility, window, coverage and observed result
Closure	Reviewer, outcome, evidence version, residual actions and lesson learned

Verification and closure

Improvement after an action is an observation. Closure also requires the configured verification period, sufficient valid data and the required reviewer decision. Missing or stale inputs make a criterion incomplete or unavailable rather than passed.

The 24-hour period and 1 percent of span threshold are demonstration criteria. Configure real criteria per case or approved template, including whether interruptions pause or restart observation. Enforce closure rules on the server. Use comparable vertical scales and exposure, capture the lesson learned, and distinguish event onset, receipt time and clock uncertainty. The concept case detection and deviation times require one consistent fixture.

Shared components and implementation sequence

Component	Used by	Key contract
AppShell and ContextHeader	All modules	Plant, search, theme, time, environment and user context
AssetPicker and ContextLink	All modules	Canonical identity and preserved investigation context
StatusBadge and DiagnosticBadge	All modules; Instrumental	Separate semantic field, text label and appropriate colour token
MetricSummary and DataTable	All modules	Units, coverage, sorting, selection and empty/error states
TrendPanel and EventTimeline	CPA, Instrumental, PLC, Downtime, RCA	Time alignment, gaps, source quality and keyboard alternative
AssetView and RelationshipView	Truss; RCA where appropriate	Verified topology versus hypothesis, selectable identities
EvidencePanel and SourceLink	All modules	Durable record reference, source time and actual access state
ReviewForm and AuditHistory	Investigation and review pages	Validation, permissions, versioning and saved outcome
AIAssessment and VerificationPanel	AI views; RCA	Authorship, evidence, criterion result and human review state

Build sequence

1. Create the semantic token layer and shell in the existing frontend. Map the existing React, TypeScript and OpenBridge components to these roles through a shared adapter where needed.
2. Build tables, trends, statuses, inspectors and evidence links using a coherent synthetic fixture. Exercise Day and Night on the same page.
3. Implement one complete path: select asset, inspect signal, review loop or device history, inspect an event, then open a linked RCA.
4. Add domain adapters and permissions, validate metric and source contracts, then replace fixture values with connected results.
5. Complete page acceptance checks and resolve the decisions on page 29 before releasing the affected live workflow.

The supplied CSS contains working colour and basic component examples. The page geometry in this guide is the shared baseline. Keep module CSS from redefining brand or semantic meanings. Backend endpoint names and library versions are outside this handoff.

Shared acceptance criteria

Use these checks during implementation review. They are acceptance requirements for the built interface; the static concepts do not demonstrate that they already pass.

ID	Check	Pass condition
G01	Theme parity	Every page renders in Day and Night with unchanged domain meaning and user context.
G02	Token consistency	Colours resolve from the supplied semantic tokens; no module brand overrides or sampled raster colours.
G03	Shared shell	One logo, icon set, module order, search label, time selector and selection treatment across the suite.
G04	Keyboard access	Primary tasks are operable by keyboard with a visible unclipped focus indicator and labelled controls.
G05	Colour independence	Status, selection and chart meaning remain understandable using labels, shapes or line styles.
G06	Data states	Loading, empty, partial, stale, bad-quality, unavailable and restricted states are distinguishable.
G07	Context continuity	Plant, identity, interval and time zone survive navigation and return to the previous view.
G08	Numerical consistency	Cards, tables, chart geometry and exports use the same dataset, units, scope and calculation version.
G09	Source traceability	Evidence actions resolve to real permitted records; missing or broken bindings remain explicit.
G10	Review integrity	Saving validates fields, records actor and time, handles concurrency and preserves audit history.
G11	AI transparency	AI output shows authorship, review state, evidence and missing coverage without fabricated confidence.
G12	Responsive clarity	Reference desktop widths, zoom and narrow layouts retain key identity, units, actions and legible content.

Visual review conditions

Review the implemented pages at 1440 by 1024, 1366 by 768 and 1920 by 1080, in both themes. Include long tags, large numbers, missing sources and many records. Compare hierarchy and density with the concepts while applying the corrections on the next page.

Do not claim complete WCAG conformance from the 94 colour checks. Keyboard, semantics, zoom, reflow, motion and actual component rendering require their own review. [6]

Module checks and concept corrections

Module	Acceptance scenario
Truss	Select an asset, inspect a real signal binding and open a linked source without losing identity or time context.
CPA	Open a loop from the ranked overview; align PV, SP, output and mode; expose the definition and coverage of each metric.
Instrumental	Open device history from a diagnostic category; retain installation identity and separate PV quality from device condition.
PLC	After scope confirmation, reconcile plan, actual, gap and context records to one period, boundary and fixture.
Downtime	Reconcile the three non-overlapping intervals to 61 minutes; an overlapping context lane does not increase that total.
RCA	A case with six valid hours against a 24-hour criterion remains incomplete and cannot close through the normal workflow.

Corrections that override the illustrations

- Truss spatial art: 50-EX-5002 must represent GearPump_Pelletizer. Replace unverified equipment geometry and routes with maintained asset information.
- Shared shell: standardize logo geometry, module icons, search wording, user display and selected navigation treatment. The examples vary in these details.
- PLC plots and records: make the 1176 t endpoint match the card; use consistent source-gap and plan-revision records between pages.
- Downtime timeline: the 10:00 to 11:00 control must either select a window or actually zoom the time axis. Do not label a selection as a completed zoom.
- Downtime recovery: reconcile the 10:37 label with the selected recovery definition and plotted data. Clarify whether the event is stoppage or reduced-rate operation.
- CPA: show the actual plotted zoom window. Reconcile overview mode compliance and detail time in AUTO to their defined windows. Use legible chart tokens for essential histogram marks.
- Instrumental: make diagnostic interval widths match their durations. Keep latest Good or Current labels separate from historical gaps. Device photographs are illustrative until matched to the installed model.
- RCA support state: use an informational treatment for Supported until a cause is verified. Normalize alarm count comparisons when before/after durations differ.
- All pages: rebuild charts from data, use a coherent fixture and preserve exact units and dates. Raster gradients, accidental alignments and example numeric shapes are not implementation specifications.

The original screen images are retained so the team can review the visual direction. These written corrections resolve their material inconsistencies without silently changing the reference set.

Decisions required for live workflows

The shared visual system and reusable components can proceed. The following decisions define the affected domain behaviour. Assign the relevant product and plant representatives during implementation; the role suggestions below do not assign an individual.

Decision	Responsible role	Needed before
PLC meaning, users and owned workflow	Product lead and module owner	Final PLC routes, fields and business logic
Plant time zone and reporting-day boundary	Operations and data owner	Daily aggregation and cross-source time comparisons
Asset identity, relationship authority and spatial model	Truss and asset-data owner	Confirmed topology and source navigation
CPA metric definitions and eligible operating periods	Control engineering and CPA owner	Ranking, thresholds and diagnostic claims
Device diagnostic aggregation and source label mapping	Instrumentation and Instrumental owner	Fleet counters and normalized status display
Alarm priority and symbol mapping	Alarm management and operations	Any source alarm colour or acknowledgement workflow
Stoppage, slowdown and recovery definitions	Operations and production accounting	Event detection and duration reconciliation
Plan version, loss baseline and missing-data policy	Planning and production accounting	Production variance and accounted loss
Case review, closure and reopen criteria	Reliability and RCA owner	Verification workflow and closure permissions
Evidence retention, review roles and audit history	Product, data and security owners	Persistent reviews and source-linked investigations

Release discipline

Version the token source and shared component library together. Record changes to colour semantics, workflow states or metric definitions in the release notes. Review theme parity when any shared component changes.

Keep operational alarm priority and source status mappings configurable by the approved site rules. Extend the suite through the shared shell and semantic tokens; add a module-specific layout only when its task requires it.

The scope decision for PLC does not block the other five module designs. The remaining domain decisions govern live behaviour and data interpretation rather than the overall Graphite and Cyan direction.

References and implementation files

External references support the accessibility criteria and diagnostic terminology. The exact Traverse colours, page layouts and proposed component dimensions are design decisions in this handoff. References checked 04 November 2026.

[1] W3C WCAG 2.2 Understanding Contrast Minimum

https://www.w3.org/WAI/WCAG22/Understanding/contrast-minimum

[2] W3C WCAG 2.2 Understanding Non text Contrast

https://www.w3.org/WAI/WCAG22/Understanding/non-text-contrast.html

[3] W3C WCAG 2.2 Understanding Use of Color

https://www.w3.org/WAI/WCAG22/Understanding/use-of-color.html

[4] NAMUR NE107 revision announcement published 18 July 2025

https://www.namur.net/en/publications/news-archive/ne107-self-monitoring-and-diagnostics-of-field-devices-has-been-revised.html

[5] Rockwell Automation and Burkert NAMUR status documentation

https://www.rockwellautomation.com/en-il/docs/add-on-profiles/partner/burkert/flowave-8098/1/flowave-8098-ditamap/online-settings-%26-diagnostics/namur-status.html

[6] Traverse Design System Foundations R01. Token source, matching CSS and numeric contrast audit prepared 03 November 2026. Included in the implementation package.

Files supplied for development

File or folder	Purpose
Guide DOCX and PDF	The same consolidated document in editable and review formats
screens	Twelve original full-resolution PNG screen concepts with descriptive names
design-system/traverse-tokens.json	Custom semantic colour schema with both themes and fixed diagnostic colours
design-system/traverse-theme.css	Matching CSS variables and basic component examples
design-system/contrast-audit files	Readable and machine-readable results for the 94 audited colour pairs
screen-manifest.json and README	Screen IDs, modes, document pages, file hashes and handoff instructions

Implement the screens as accessible native UI components. The PNGs remain reference artwork. Use the document for behaviour and precedence, the JSON for exact colour values and the service contracts for live data.